www.tntextbooks.in

1. Differentiation

The process of maturation of meristematic cells to specific types of cells performing specific functions is called **differentiation**.

2. Dedifferentiation

The living differentiated cells which had lost capacity to divide, regain the capacity to divide under certain conditions. Hence, dedifferentiation is the regaining of the ability of cell division by the differentiated cells. Example: Interfascicular cambium and Vascular cambium.

3. Redifferentiation

Differentiated cells, after multiplication again lose the ability to divide and mature to perform specific functions. This is called **redifferentiation** (Figure 15.9). Example: Secondary xylem and Secondary phloem.

4. Plasticity

Plants follow different pathways in response to environment or phases of life to form different kinds of structures.

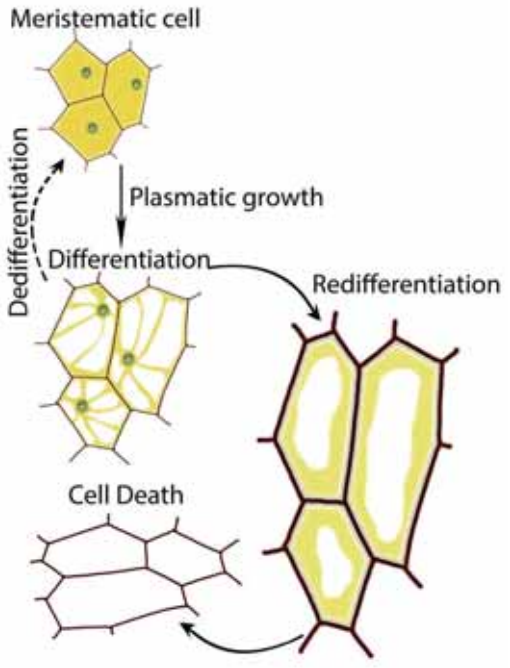


Figure 15.9: Sequences of developmental process in a plant cell

This ability is called **plasticity**. Example: Heterophylly in cotton and coriander. In such plants, the leaves of the juvenile plant are different in shape from those in mature plants. On the other hand, the difference in shapes of leaves produced in air and those produced in water in buttercup also represent the heterophyllous development due to the environment. This phenomenon of heterophylly is an example of plasticity.

15.2 Plant Growth Regulators

CRK9N8

Plant Growth Regulators (chemical messenger) are defined as organic substances which are synthesized in minute quantities in one part of the plant body and transported to another part where they influence specific physiological processes. Five major groups of hormones *viz.*, auxins, gibberellins, cytokinins, ethylene and abscisic acid are presently known to coordinate and regulate growth and development in plants. The term **phytohormones** is implied to those chemical substances which are synthesized by plants and thus, naturally occurring. On the other hand, there are several manufactured chemicals which often resemble the hormones in physiological action and even in molecular structure. Recently, another two groups, the brassinosteroids and polyamines were also known to behave like hormones.

1. Plant growth regulators – classification

Plant Growth Regulators are classified as natural and synthetic based on their